Solving the Model Gap

Why does cheaper, better silicon not win by itself?

AMD-Sourcing the Engineering Cycles to Close NVIDIA's Physical-AI Lead -- A proposal

1 · The Paradox We Keep Hitting in Sales

We Win the Spec Sheet and Lose the Deal

The customer never disputes the economics — he refuses because the decision was made above the silicon.

AMD hardware wins on economics

- **~50% lower capex** and **~30% lower opex** vs the incumbent (on the right criteria)
- Competitive silicon: MI300X **5.3 TB/s** HBM3, Ryzen AI edge, Kria deterministic control

> On price-performance, we may have/reach a real, quantifiable advantage.

…yet the application developer still says "no"

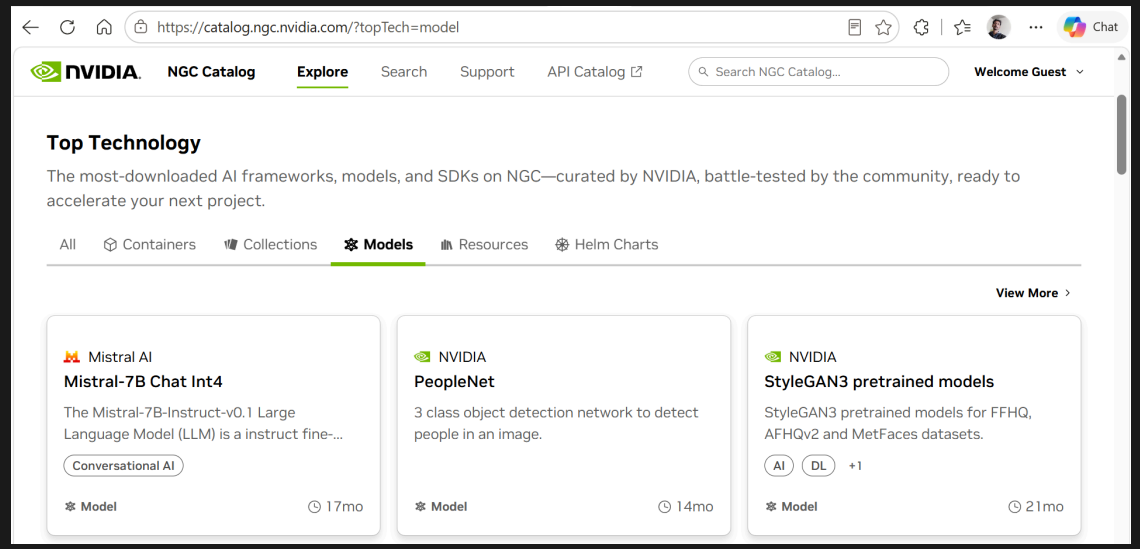
- His product is already built **around NVIDIA's models and pipelines**
- Switching means **re-engineering the whole stack** — a cost *he* has to pay
- The better chip **never gets a fair evaluation**

> We lose the deal **before** hardware economics enter the conversation.

> **The barrier is not the silicon — it's the ecosystem the customer has already built on top of NVIDIA.**

2 · NVIDIA's Moat Is Software, Not the Chip

A Decade of Models Turned Into a Vertically Integrated Stack



NGC Catalog — years of curated, prepackaged, ready-to-run models

The lock-in is measurable

- **~80% ecosystem, ~20% hardware** — NVIDIA's own default status
- **2M+ robotics developers**, **4M CUDA developers**, **7+ year** head start
- **Isaac Sim** leads robotics job postings — **50% more** than the closest competitor
- **1,228 VLA papers** at ICLR 2026 → **zero** mention ROCm

Models → TensorRT → Isaac ROS → cuRobo → Isaac Sim → Cosmos — where each layer locks in the next — and locks AMD out.

3 · The Switching Cost Is the Real Barrier

The Customer Would Take Cheaper Hardware — but He Pays the Switching Bill

Every engineering cycle spent re-hosting on AMD is capex the customer must justify before he saves a cent.

Why the customer stays put

- Years of infra, pipelines, and applications built on NVIDIA models
- Rebuilding fleet/deployment infra: **6–18 engineer-months per deployment**
- **5–7 deployment artifacts per model** to re-target hardware
- Even hardware-competitive rivals face a **12–24 month** software catch-up

The burden lands hardest where we want to win

Small / mid-size robotics companies
every engineering cycle is capex they cannot spare

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They keep buying the more expensive chip — because **switching costs more than it saves**

> **This is the soft-power gap — AMD is absent from every on-ramp above the compute layer, and it shows up directly in sales.**

4 · The Internal Problem: ROI Can't Be Justified Overnight

The Gap Is Known and Winnable — But ROI can not be defended

How AMD is structured

- Many business units — each funded by its **own sales / engineering output**
- Every unit must **defend the returns** on what it takes on

Why no single unit picks this up

- Model training is **iterative and continual** — not a one-shot deliverable
- A **multi-year ecosystem gap** cannot be closed overnight
- ROI in a fixed window is **hard to measure and harder to justify**

The Deadlock

Gap is **large** & the payoff is **long-horizon**

↓

No BU can **book near-term ROI**

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So **nobody starts** — and the gap widens

A multi-year, iterative gap cannot be booked as near-term ROI — so every business unit is rational to pass.

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5 · The Insight: The Target Is Already Known

When the models and target numbers are published, closing the gap becomes bounded, measurable work.

Unlike open-ended research, we are not searching

- The **models are named** and mostly **open** (NGC is a public map)
- The **target performance is published** — we know exactly what "done" looks like
- This is a **replication problem**, not an invention problem
- The **replication problem**, can be solved at scale

> Producing a replica of a known target is **bounded, measurable work** — the ROI objection shrinks.

Only two pieces are missing

Compute — we already have it

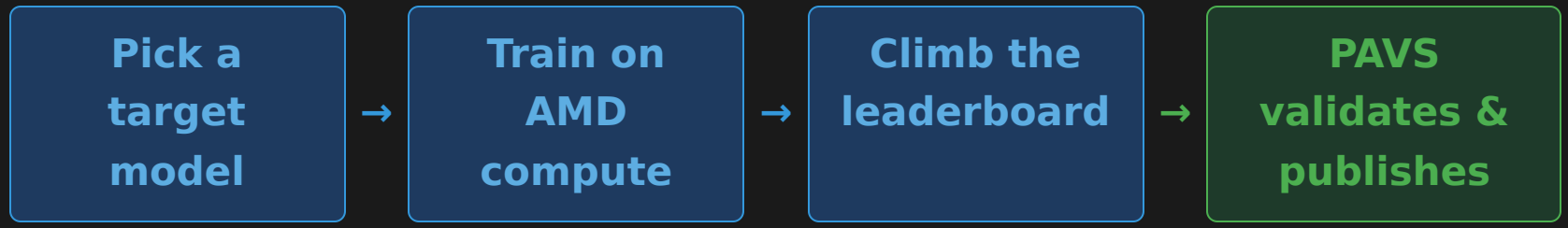
AMD-wide clusters (ALOLA & others)

Data — the real problem to solve

synthetic generation · **pooled + curated open data** · **semi-supervised for the task** · **pseudo-labels from large OSS models** · & more

> Known target + available compute → the task collapses to **solving the data problem at scale**.

6 · The Proposal: An AMD-Wide Model Training Challenge



The mechanics

- A **leaderboard per model**
- A **webpage** tracking the top teams and models
- Each **new Update notifies every participant** — winner + solution shared — keeping the field competitive
- **Top 3–4 teams** get to **publish their work**

Why a challenge, not a hire

- Taps **latent enthusiast engineering** across all of AMD
- Converts a **multi-year gap** into **many parallel short sprints**
- No single BU carries the bill — the org **contributes cycles**

7 · Challenges Have Closed Big Capability Gaps Before

We're not guessing this works — a scored, public target has repeatedly closed gaps no single roadmap could.

ImageNet Challenge

- A **public leaderboard** on a known target
- Sparked **AlexNet** and the deep-learning era
- A benchmark did what no single lab's roadmap could

DARPA Grand Challenge

- A **prize + leaderboard** for autonomous driving
- Year 1: nobody finished. Year 2: **multiple teams did**
- Seeded the entire self-driving industry

Kaggle / Netflix Prize

- **Crowdsourced** solutions to a fixed, scored target
- Outperformed in-house teams at a **fraction of the cost**
- Recognition — not salary — drove the effort

"The right *challenge challenges* the right engineer — and produces great engineering."

8 · The Target Set Is Open — and Achievable

One AMD alternative per NGC model — the 30 models a customer has running today and can't move off NVIDIA for.

Tier 1 + 2 — the real "can't move it" set (30 models)

Tier	Function	Models
Tier 1 perception	Detection	TrafficCamNet, PeopleNet / PeopleNet-AMR
	Recognition/Re-ID	LPRNet + LPDNet, VehicleMakeNet / VehicleTypeNet, ReIdentificationNet
	Segmentation	PeopleSemSegNet / AMR
	Retail	Object Detection, Object Recognition
	Behavior/Pose	Action Recognition Net, BodyPoseNet / PoseClassificationNet
	OCR	OCDNet / OCRNet
Tier 2 robotics	Stereo/3D	Bi3D, ESS, PointPillarNet, Transfusion
	Pose	FoundationPose
	Segmentation	CitySemSegFormer
	Backbones	TAO Pretrained (Classification, DetectNet V2, EfficientDet, Instance Seg, Semantic Seg)

> full 4-tier, 48-model table with rationale is presented in **Appendix A**

The pieces already exist — what's missing is organized, incentivized effort pointed at AMD silicon.

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9 · Incentive Design — Everyone Wins

A Challenge Only Works if Every Party Wins by Playing — Here, Three Do

The engineer, PAVS, and AMD each get something they want — which is why the mechanism sustains itself.

Participant (Engineer)

- **Recognition** — publish work, top-team webpage
- A **countable contribution to AMD** for their record
- An **extra purpose & goal** alongside the day job
- Enthusiast hours become **productive output**

Organizer (PAVS)

- Becomes the **customer gateway** to physical AI across AMD
- Reproducing models **builds working ties** across many teams
- A **niche — but the lead player** in that niche
- **Bridges the islands** of physical AI at AMD

AMD (Org)

- The **ecosystem gap closes fast** — target is known
- A growing **pool of curated datasets** — one team's data & models **catalyze the next**
- **Idle compute** (ALOLA & others) gets **popularized**
- A pipeline of **market-ready, AMD-native models**
- Engagement without a **single-BU budget line**

10 · PAVS as the Customer Gateway to Physical AI

Organizing the Challenge Isn't Overhead — It's How PAVS Earns Its Position

Reproducing and routing every winning model is exactly what makes PAVS the gateway physical AI flows through.

What PAVS commits to

- Send an **org-wide call** for the right technical solutions
- The right solution that reaches us → **we help take it to market**
- **Reproduce, evaluate, and certify** each model + its performance
- Curate the leaderboard and the **model webpage**

What that makes PAVS

The gateway — every physical-AI solution enters through PAVS

↓

The bridge — connecting the islands of physical AI across AMD

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The lead player in a well-defined niche

The organizing work is not overhead — it's how PAVS earns the relationships and the position to route physical AI to market on AMD.

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11 · An Enormous Market — The Prize Is Worth the Effort

A drop-in AMD replica per NGC model turns a costly migration into a seamless switch — and unlocks the customers already locked in.

$200B

AMD's own projection for the physical-AI silicon market by 2035

$22.2B

robotics startup funding in 2025 — up 69% YoY; the buyers of tomorrow are forming now

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of 1,228 VLA papers at ICLR 2026 mention ROCm — the on-ramp is wide open to claim

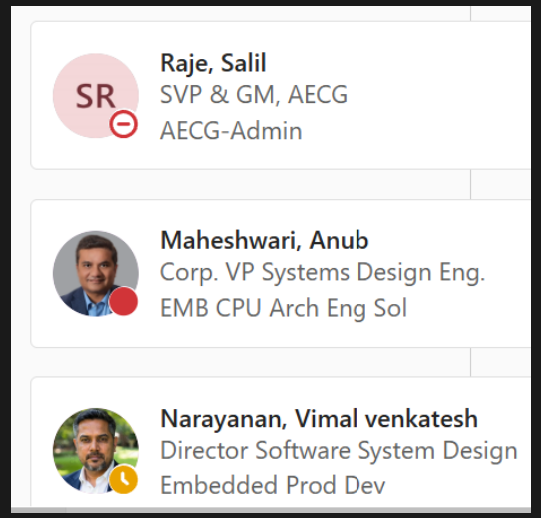
> **Make the transfer seamless and AMD's price-performance is finally evaluated — the cheaper stack wins the deals it loses today.**

The Ask

The plan is already staffed — we only need the mandate to start

What we need from leadership

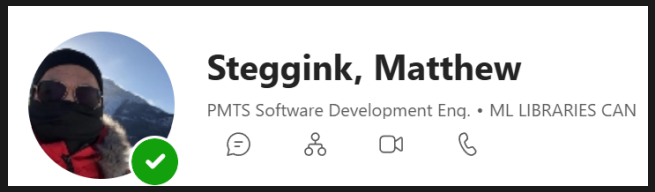
> An **org-wide email** endorsing the challenge — the signal that gives the effort **good traction**.



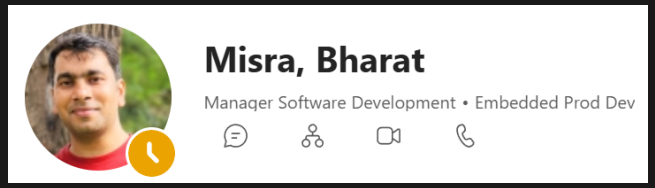
The push

Already staffed & committed — execution is ready

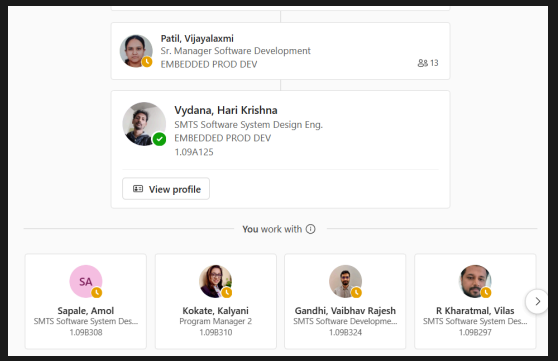
Compute — grants cluster access (ALOLA & AMD-wide)



Per-model GitHub — creates the per-model repos



Team to drive the work



Known target · staffed team · available compute — all that's missing is the go-ahead.

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Appendix A1 · Tier 1 — Production Perception Stack (NVIDIA Metropolis/DeepStream)

Highest priority — most commonly deployed NGC models a customer runs today and cannot carry over to AMD.

NVIDIA model	Function	Why it matters
TrafficCamNet	Vehicle detection (traffic cameras)	Core Metropolis/smart-city SKU
PeopleNet / PeopleNet-AMR	Person detection	Most widely deployed DeepStream model
License Plate Recognition (LPRNet) + LPDNet	Plate detection + recognition pair	Sold as a pair, tolling/parking/security
VehicleMakeNet / VehicleTypeNet	Vehicle attribute classification	Paired with TrafficCamNet in the same pipelines
ReIdentificationNet (+ Transformer variant)	Person re-identification across cameras	Retail/security analytics
PeopleSemSegNet / PeopleSemSegNet AMR	Person segmentation	Privacy-masking, robotics safety zones
Retail Object Detection / Retail Object Recognition	Shelf/product detection	Retail analytics vertical
Action Recognition Net	Action/behavior classification	Security, workplace safety
BodyPoseNet / PoseClassificationNet	Human pose estimation	Robotics, ergonomics, fitness
Optical Character Detection/Recognition (OCDNet/OCRNet)	Scene text OCR	Logistics, document/label reading

Appendix A2 · Tier 2 — Robotics/Perception Building Blocks (Physical AI)

The models Isaac ROS perception pipelines are built on — closest to AMD-ROS/AARTS.

NVIDIA model	Function
Bi3D Proximity Segmentation	Stereo-based obstacle proximity
ESS DNN Stereo Disparity	Stereo depth
FoundationPose	6-DoF object pose estimation/tracking
PointPillarNet	LiDAR 3D object detection
Transfusion for 3D Object Detection	Camera+LiDAR fusion detection
CitySemSegFormer	Semantic segmentation (autonomy/city-scale)
TAO Pretrained Classification/DetectNet V2/EfficientDet/Instance Seg/Semantic Seg	Generic backbones customers fine-tune on top of — AMD needs *some* pretrained backbone family here or customers can't even start a TAO-style workflow on AMD

> Tier 1 + Tier 2 = **30 models** — the real "customer has this running on NVIDIA today and can't move it" set.

Appendix A3 · Tier 3 — Biometric / Human-Sensing (Smaller but Sticky Niche)

Matters mainly if AMD is also chasing the biometric/kiosk vertical.

NVIDIA model	Function
FaceNet	Face detection
FaceDetectIR	Face detection (infrared)
GazeNet (Gaze Estimation)	Gaze tracking
EmotionNet	Emotion classification
GestureNet	Hand/body gesture recognition
HeartRateNet	Remote heart-rate estimation
Facial Landmark Estimator (FPENet)	Facial landmark detection
Pre-trained Deformable DETR NvImageNet Weights	Detection backbone
Pre-trained FAN based NVImageNet Classification Weights	Classification backbone
Pre-trained GCViT NVImageNet Classification Weights	Classification backbone
Visual ChangeNet Classification	Change detection classification
NVIDIA Optical Inspection Model	Industrial visual inspection

Appendix A4 · Tier 4 — Enterprise Vision & HPC/Climate (Lowest Priority)

Matters mainly if AMD is also chasing the enterprise-vision or HPC-climate vertical.

NVIDIA model	Function	Why lower priority
PCB Defect Classification Model	Industrial inspection	NVIDIA AI Enterprise EULA; small footprint
PeopleSemSegFormer Model Card	Person segmentation	NVIDIA AI Enterprise EULA; small footprint
Pre-trained DINO NvImageNet Weights	Classification backbone	NVIDIA AI Enterprise EULA; small footprint
PhysicsNeMo Checkpoints: AFNO_DX_FI-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: AFNO_DX_SR-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: AFNO_DX_TP-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: AFNO_DX_WG-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance
PhysicsNeMo Checkpoints: DLESyM-V1-ERA5	Weather forecasting	NVIDIA Community Model License; HPC/climate-only relevance

48 models across 4 tiers, out of 125 scanned on NGC — the other 77 are either already portable (49) or unresolved catalog entries.